

SERPO: SENTENCE-LEVEL ENTROPY REWARDED POLICY OPTIMIZATION

Author(s) Name(s)

Author Affiliation(s)

ABSTRACT

The abstract should appear at the top of the left-hand column of text, about 0.5 inch (12 mm) below the title area and no more than 3.125 inches (80 mm) in length. Leave a 0.5 inch (12 mm) space between the end of the abstract and the beginning of the main text. The abstract should contain about 100 to 150 words, and should be identical to the abstract text submitted electronically along with the paper cover sheet. All manuscripts must be in English, printed in black ink.

Index Terms— One, two, three, four, five

1. INTRODUCTION

Group-Based Policy Optimization (GRPO) [1] suffers from coarse-grained credit assignment, applying uniform rewards across entire token sequences based on the final outcome, without capturing the varying contributions of different sequence components to response quality. This approach ignores the hierarchical nature of language generation, where semantic units contribute differently to overall quality.

We propose sentence-level reward assignment (equivalently advantage) as an intermediate granularity between token-level and sequence-level approaches. Our method assigns constant rewards to sub-sentences within responses, creating structured reward signals that better reflect language’s compositional nature. This approach offers key advantages: improved reward normalization that allows important reasoning steps to receive appropriate credit, semantic coherence aligned with human text evaluation patterns, preservation of reasoning chain integrity, and human-aligned evaluation that mirrors how people naturally assess complete thoughts rather than individual words.

This sentence-level approach offers several benefits: 1. Improved Normalization: By operating at the sentence level, we achieve better reward normalization across sentences, allowing important reasoning steps to receive appropriate credit. 2. Semantic Coherence: Sentence-level rewards align more closely with how humans evaluate text quality, where coherent thoughts and complete ideas are assessed as units rather than individual words. 3. Reasoning Chain Preservation: Important sequences in multi-step reasoning maintain

their structural integrity, as the full sentence guides the reasoning process rather than fragmented token-level signals. 4. Human-Aligned Evaluation: This approach replicates human evaluation patterns, where we naturally assign scores to sentences or paragraphs rather than individual words, making the reward signal more interpretable

2. RELATED WORK

Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a dominant paradigm for enhancing reasoning capabilities in large language models. Group Relative Policy Optimization (GRPO) [1] has become a widely adopted approach, eliminating the need for separate value functions by using group-average rewards as baselines. While effective, GRPO applies uniform advantages across all tokens within a sequence, creating a fundamental credit assignment problem where all tokens receive identical learning signals regardless of their contribution to the final outcome.

Recent work has revealed critical insights into the role of individual tokens during RLVR training. [2] demonstrated that only a minority of tokens ($\sim 20\%$) exhibit high entropy and act as “forking points” that determine reasoning trajectories. Their analysis showed that these high-entropy tokens are responsible for nearly all performance gains in RLVR:

$$H_t := - \sum_{j=1}^V p_{t,j} \log p_{t,j} \quad (1)$$

where π_θ is the policy parameterized by θ (the model parameters), $p_{t,j} = \pi_\theta(j|q, o_{<t})$ represents the probability distribution over vocabulary V at position t , q denotes the input query, and $o_{<t}$ represents the previously generated tokens. They found that restricting policy gradients to only the top 20% highest-entropy tokens maintains or exceeds full-gradient performance.

[3] introduced Dynamic Entropy Weighting through Group Token Policy Optimization (GTPO), assigning entropy-weighted rewards to individual tokens:

$$\tilde{r}_{i,t} = \alpha_1 r_i + \alpha_2 \frac{H_{i,t}}{\sum_{k=1}^n H_{k,t}} \cdot d_t \quad (2)$$

where $H_{i,t}$ is the entropy when generating token $o_{i,t}$, d_t is the number of active sequences at timestep t , and $\alpha_1, \alpha_2 > 0$

Thanks to XYZ agency for funding.

are weighting coefficients with $\alpha_1 + \alpha_2 = 1$. This approach addresses the coarse-grained credit assignment problem by providing fine-grained, token-specific reward signals.

[4] identified entropy collapse as a fundamental limitation in RLVR, establishing an empirical relationship between policy entropy H and performance R :

$$R = -a \exp(H) + b \quad (3)$$

where a, b are fitting coefficients that reflect the internal characteristics of the policy and data distribution. This relationship suggests that performance is “traded” from entropy in a predictable manner, with policy performance ceiling determined when $H = 0$, yielding $R = -a + b$. They proposed Clip-Cov and KL-Cov methods to control entropy by restricting updates of high-covariance tokens.

[5] further explored the entropy-exploration connection, introducing entropy-based advantage shaping:

$$A_t^{\text{shaped}} = A_t + \psi(H_t) \quad (4)$$

$$\psi(H_t) = \min \left(\alpha \cdot H_t^{\text{detach}}, \frac{|A_t|}{\kappa} \right) \quad (5)$$

where A_t is the original advantage, $\alpha > 0$ is a scaling coefficient, $\kappa > 1$ controls the clipping threshold, and the entropy term H_t^{detach} is gradient-detached to preserve original optimization dynamics while encouraging exploratory reasoning behaviors.

The MiniMax team [6] proposed CISPO (Clipped Importance Sampling Policy Optimization), which addresses entropy management by clipping importance sampling weights rather than token updates. Their approach explicitly avoids dropping tokens, even those associated with large updates, while inherently maintaining entropy within a reasonable range to ensure stable exploration. CISPO demonstrates improved efficiency compared to GRPO and DAPO, achieving 2x speedup in controlled studies while maintaining training stability through importance weight constraints rather than traditional trust region methods.

[7] introduced EDGE-GRPO, which addresses the advantage collapse problem in GRPO through Entropy-Driven Advantage (EDA). Their approach recognizes that policy entropy exhibits miscalibration at the sample level—incorrect responses do not necessarily indicate high uncertainty, and correct responses may display relatively high entropy. They propose entropy-driven advantage scaling:

$$\hat{A}_i = \frac{A_i}{\hat{P}_i} \quad (6)$$

where $\hat{P}_i = \frac{P_i}{\text{mean}(\{P_1, P_2, \dots, P_G\})}$ is the normalized policy entropy for response i , and P_i is the average token-level entropy across the response. This scaling assigns higher advantages to confident correct responses while penalizing overcon-

fident incorrect responses, addressing advantage collapse by increasing signal diversity.

While most work focuses on token-level or sequence-level rewards, some research has explored intermediate granularities. [8] in their MAGIC framework investigated multi-stage training and advantage masking for truncated responses, though their focus remained primarily on sequence-level optimization.

[9] proposed Group-relative Trajectory-based Policy Optimization (GTPO), addressing gradient conflicts in GRPO by identifying “conflict tokens” that appear across completions with opposing rewards. Their conflict-aware gradient correction applies selective masking:

$$\lambda_{i,t} = \begin{cases} 1 & \text{if } M_{i,t} = 0 \\ 0 & \text{if } M_{i,t} = 1 \text{ and } A_i < 0 \\ 2 & \text{if } M_{i,t} = 1 \text{ and } A_i > 0 \end{cases} \quad (7)$$

where $M_{i,t}$ is a binary mask identifying conflict positions, A_i represents the normalized advantage for completion i , and $\lambda_{i,t}$ is the token-level weight applied during gradient updates. GTPO eliminates KL regularization in favor of entropy-based policy regularization to prevent policy collapse.

Recent work has also explored different levels of reward granularity in reasoning tasks. However, most approaches operate at either the token level (prone to noise) or the sequence level (suffering from coarse credit assignment), leaving a gap for semantically meaningful intermediate approaches that balance stability with semantic coherence.

3. PROBLEM FORMULATION

As previously done in [10], we introduce the step entropy, that will measure the uncertainty of the model of each step in the reasoning. Given a chain-of-thought (CoT) C produced by a Large Reasoning Model (LRM), we first split it as a series of distinct steps $C = (S_1, S_2, \dots, S_N)$, where each step S_i is a sequence of tokens, $S_i = (t_{i,1}, t_{i,2}, \dots, t_{i,M_i})$. And for each token $t_{i,j}$, the model outputs a probability distribution $p(\cdot | c_{i,j})$ over the vocabulary V , where the context $c_{i,j}$ includes the input prompt together with all previously generated tokens.

The per-token entropy of this distribution, which represents the model’s uncertainty at that generation step, is calculated using the standard Shannon entropy formula

$$H(t_{i,j} | c_{i,j}) = - \sum_{w \in V} p(w | c_{i,j}) \log_2 p(w | c_{i,j}) \quad (8)$$

We define the step entropy $H(S_i | S_{<i})$ as the sum of token-level entropy across all tokens within that step:

$$H(S_i | S_{<i}) = \sum_{j=1}^{M_i} H(t_{i,j} | c_{i,j}) \quad (9)$$

The sequence entropy is interpreted analogously to the per-token entropy: high entropy implies greater uncertainty in generation, while low entropy indicates more confidence. Let S_j be a sequence in this CoT and A the final answer, We consider the conditional mutual information, which measures how much S_j uniquely contributes to explaining or predicting A beyond what is captured by the remaining sentences :

$$\begin{aligned} I(S_j; A | \bar{S}_j) &= I(S_j; A | S_1, \dots, S_{j-1}, S_{j+1}, \dots, S_N) \\ &= H(S_j | \bar{S}_j) - H(S_j | \bar{S}_j, A) \\ &\leq H(S_j | \bar{S}_j) = H(S_j | S_{<j}, S_{>j}) \\ &= H(S_j | S_{<j}) - I(S_j; S_{>j} | S_{<j}) \end{aligned}$$

We get that,

$$I(S_j; A | \bar{S}_j) \leq H(S_j | S_{<j}) \quad (10)$$

, as $I(S_j; S_{>j} | S_{<j}) \geq 0$. And thus, lower entropy sequences contributes less to the final answer. Our main idea is that these sequences should receive lower rewards, so the model learns to produce sentences that contribute more highly to final answer A , which could potentially lead to better problem resolution.

4. SENTENCE-LEVEL ENTROPY REWARDED POLICY OPTIMIZATION - SERPO

Determining whether a value is low or high is relative, as we need a reference point for comparison. To address this, we compare sentences against each other by assigning negative rewards to those below the mean sequence entropy (considered low entropy). Furthermore, to preserve the core behavior of previous used SOTA algorithms like GRPO/DAPO, which use constant rewards for each rollout, and to control the effect of this entropy-based reward, we add it directly to the advantage rather than modifying the fundamental approach of GRPO. Consequently, we also normalize this advantage by standard deviation (though this doesn't yield probabilistically normalized results since the sequences are interdependent). For each prompt i and for each sequence j in the rollout of i :

$$\hat{B}_{i,j} = \hat{A}_i + \lambda Z_{i,j} \quad (11)$$

, Where $\hat{A}_i$ is the usual advantage of GRPO. λ is the parameter that controls the effect of this per-sequence reward. And $Z_{i,j}$ is contribution of the per-sequence reward.

Let $E_{i,j}$ denote the detached entropy of sequence $S_{i,j}$, the j -th sequence in rollout i . To avoid bias due to sequence length, we normalize each sequence entropy by its length. We define $H_{i,j} = \frac{1}{|S_{i,j}|} E_{i,j}$, where $|S_{i,j}|$ is the number of valid tokens in the sequence.

The entropy contribution for sequence j in prompt i is:

$$Z_{i,j} = \frac{H_{i,j} - \bar{H}_i}{\sigma_{H_i}} \quad (12)$$

where $\bar{H}_i = \frac{1}{n_i} \sum_{k=1}^{n_i} H_{i,k}$ and

$$\sigma_{H_i} = \sqrt{\frac{1}{n_i-1} \sum_{k=1}^{n_i} (H_{i,k} - \bar{H}_i)^2}.$$

This approach introduces a new advantage function that remains constant across all tokens within each sequence of a generation—unlike previous methods where the advantage was constant across the entire generation. The goal is to subtly enhance specific behaviors and guide gradient updates toward desired directions, without fully altering the underlying rewards or advantage values.

Another technical decision was to restrict this behavior to the "thinking" chain-of-thought (CoT) segments, specifically between the `<think>` and `</think>` tags. We assume the model should be confident in each token generated during the answering phase, following the thinking part in the CoT. Therefore, no additional advantage is applied in that segment, effectively setting it to zero. This aligns well with our use of normalization: an advantage > 0 indicates desired behavior, $= 0$ reflects neutrality, and < 0 signals undesired behavior.

5. EXPERIMENTS

In order to validate the SERPO approach, we conduct the following experiments. In most experiments, we use distilled models as we observe this follows the approach used in most large-scale RL pipelines: SFT on chain-of-thought data followed by RL fine-tuning. The advantage of these models is that they already produce thinking chains of thought and thinking tags that we need to compute our reward/advantage functions. Another observation is that they have intrinsically separated steps. We split each rollout using the `\n\n` delimiter (inherent behavior in most distilled models) and keep only the content between `<think>` `</think>` tags. This requires a change in how the reward is managed by the reward manager.

We use a specialized fork of the VERL project and employ two distinct experimental setups, both utilizing a curated subset of the POLARIS [11] dataset by retaining only difficulty levels 3, 4, and 5 to achieve faster convergence signals. We also incorporate rejection sampling and implement a 5-step learning rate warmup with $\text{lr} = 3 \times 10^{-6}$ for the same purpose. Our configuration includes a reasoning length of 16K tokens, group size $G = 8$, batch size = 128, and PPO mini-batch size = 32. In the first experimental setup, we applied no KL divergence regularization in both vanilla GRPO and our custom algorithm SERPO, with `clip_ratio` = 0.2. In the second setup, we added `clip_ratio_high` = 0.28 and `clip_ratio_low` = 0.2 to replicate DAPO [12] performance and to examine whether this entropy advantage to examine whether this reward serves merely to maintain higher exploration or actually helps guide the reasoning path toward better solutions by promoting more effective steps. In both setups, we used two hyperparameter λ values to control how the sequence entropy advantage addition contributes to the main

advantage function. Two values are used in most runs: 0.01 and 0.001, to limit the contribution of this per-sequence entropy advantage and not fully alter the initial RL dynamics.

Adding an entropy-dependent reward to the advantage calculation during training could potentially alter the usual dynamics of RL usual algorithms. However, this approach preserves most of the original dynamics by using the fixed reference policy $\pi_{\theta_{\text{old}}}$ to compute the entropy term. Since $\pi_{\theta_{\text{old}}}$ remains constant during each cycle of mini-batch updates, the entropy reward does not depend on the current parameters θ_t at step t within each update cycle. This design choice, combined with the mini-batch structure used in PPO-like [13] algorithms, allows the model to maintain essentially the same training dynamics as the original RL algorithm.

6. MAJOR HEADINGS

Major headings, for example, "1. Introduction", should appear in all capital letters, bold face if possible, centered in the column, with one blank line before, and one blank line after. Use a period (".") after the heading number, not a colon.

6.1. Subheadings

Subheadings should appear in lower case (initial word capitalized) in boldface. They should start at the left margin on a separate line.

6.1.1. Sub-subheadings

Sub-subheadings, as in this paragraph, are discouraged. However, if you must use them, they should appear in lower case (initial word capitalized) and start at the left margin on a separate line, with paragraph text beginning on the following line. They should be in italics.

7. PRINTING YOUR PAPER

Print your properly formatted text on high-quality, 8.5 x 11-inch white printer paper. A4 paper is also acceptable, but please leave the extra 0.5 inch (12 mm) empty at the BOTTOM of the page and follow the top and left margins as specified. If the last page of your paper is only partially filled, arrange the columns so that they are evenly balanced if possible, rather than having one long column.

In LaTeX, to start a new column (but not a new page) and help balance the last-page column lengths, you can use the command "`\pagebreak`" as demonstrated on this page (see the LaTeX source below).

8. PAGE NUMBERING

Please do **not** paginate your paper. Page numbers, session numbers, and conference identification will be inserted when

the paper is included in the proceedings.

9. ILLUSTRATIONS, GRAPHS, AND PHOTOGRAPHS

Illustrations must appear within the designated margins. They may span the two columns. If possible, position illustrations at the top of columns, rather than in the middle or at the bottom. Caption and number every illustration. All halftone illustrations must be clear black and white prints. Colors may be used, but they should be selected so as to be readable when printed on a black-only printer.

Since there are many ways, often incompatible, of including images (e.g., with experimental results) in a LaTeX document, below is an example of how to do this [?].

10. FOOTNOTES

Use footnotes sparingly (or not at all!) and place them at the bottom of the column on the page on which they are referenced. Use Times 9-point type, single-spaced. To help your readers, avoid using footnotes altogether and include necessary peripheral observations in the text (within parentheses, if you prefer, as in this sentence).

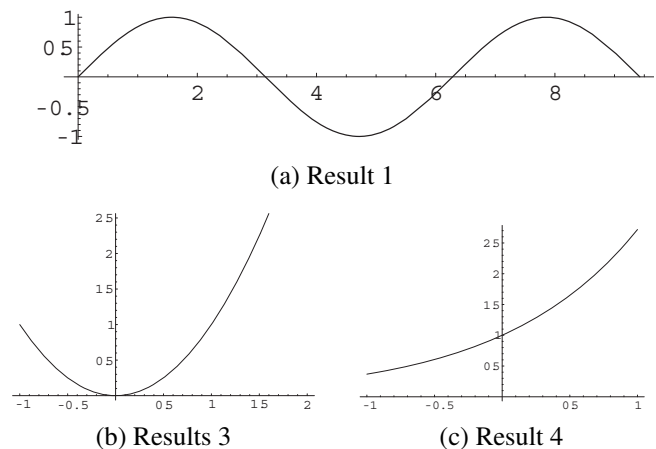


Fig. 1. Example of placing a figure with experimental results.

11. COPYRIGHT FORMS

You must submit your fully completed, signed IEEE electronic copyright release form when you submit your paper. We **must** have this form before your paper can be published in the proceedings.

12. RELATION TO PRIOR WORK

The text of the paper should contain discussions on how the paper’s contributions are related to prior work in the field. It is important to put new work in context, to give credit to foundational work, and to provide details associated with the previous work that have appeared in the literature. This discussion may be a separate, numbered section or it may appear elsewhere in the body of the manuscript, but it must be present.

You should differentiate what is new and how your work expands on or takes a different path from the prior studies. An example might read something to the effect: “The work presented here has focused on the formulation of the ABC algorithm, which takes advantage of non-uniform time-frequency domain analysis of data. The work by Smith and Cohen [?] considers only fixed time-domain analysis and the work by Jones et al [?] takes a different approach based on fixed frequency partitioning. While the present study is related to recent approaches in time-frequency analysis [3-5], it capitalizes on a new feature space, which was not considered in these earlier studies.”

13. REFERENCES

List and number all bibliographical references at the end of the paper. The references can be numbered in alphabetic order or in order of appearance in the document. When referring to them in the text, type the corresponding reference number in square brackets as shown at the end of this sentence [?]. An additional final page (the fifth page, in most cases) is allowed, but must contain only references to the prior literature.

14. REFERENCES

- [1] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo, “Deepseekmath: Pushing the limits of mathematical reasoning in open language models,” *arXiv preprint arXiv:2402.03300v2*, 2024.
- [2] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin, “Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning,” *arXiv preprint arXiv:2506.01939*, 2025.
- [3] Hongze Tan and Jianfei Pan, “Gtpo and grpo-s: Token and sequence-level reward shaping with policy entropy,” *arXiv preprint arXiv:2508.04349*, 2025.
- [4] Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, Zhiyuan Liu, Hao Peng, Lei Bai, Wanli Ouyang, Yu Cheng, Bowen Zhou, and Ning Ding, “The entropy mechanism of reinforcement learning for reasoning language models,” *arXiv preprint arXiv:2505.22617*, 2025.
- [5] Daixuan Cheng, Shaohan Huang, Xuekai Zhu, Bo Dai, Wayne Xin Zhao, Zhenliang Zhang, and Furu Wei, “Reasoning with exploration: An entropy perspective,” *arXiv preprint arXiv:2506.14758*, 2025.
- [6] MiniMax, “Minimax-m1: Scaling test-time compute efficiently with lightning attention,” *arXiv preprint arXiv:2506.13585*, 2025.
- [7] Xingjian Zhang, Siwei Wen, Wenjun Wu, and Lei Huang, “Edge-grpo: Entropy-driven grpo with guided error correction for advantage diversity,” *arXiv preprint*, 2025.
- [8] Jujie He, Jiacai Liu, Chris Yuhao Liu, Rui Yan, Chaojie Wang, Peng Cheng, Xiaoyu Zhang, Fuxiang Zhang, Jiacheng Xu, Wei Shen, Siyuan Li, Liang Zeng, Tianwen Wei, Cheng Cheng, Bo An, Yang Liu, and Yahui

Zhou, “Skywork open reasoner 1 technical report,” *arXiv preprint arXiv:2505.22312*, 2025.

- [9] Marco Simoni, Aleksandar Fontana, Giulio Rossolini, and Andrea Saracino, “Gtpo: Trajectory-based policy optimization in large language models,” *arXiv preprint arXiv:2508.03772*, 2025.
- [10] Zeju Li, Jianyuan Zhong, Ziyang Zheng, Xiangyu Wen, Zhijian Xu, Yingying Cheng, Fan Zhang, and Qiang Xu, “Compressing chain-of-thought in llms via step entropy,” *arXiv preprint arXiv:2508.03346*, 2025.
- [11] Chenxin An, Zhihui Xie, Xiaonan Li, Lei Li, Jun Zhang, Shansan Gong, Ming Zhong, Jingjing Xu, Xipeng Qiu, Mingxuan Wang, and Lingpeng Kong, “Polaris: A post-training recipe for scaling reinforcement learning on advanced reasoning models,” 2025.
- [12] ByteDance Seed, “Dapo: An open-source llm reinforcement learning system at scale,” *arXiv preprint arXiv:2503.14476*, 2025.
- [13] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347v2*, 2017.